

JOHN HU

Electrical Engineering Student at Duke University

Expected graduation: 2028 GPA: 3.633



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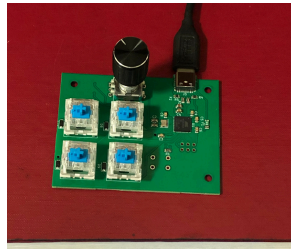
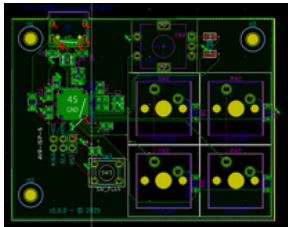
Experience

October 2025 - November 2025

MECHANICAL KEYBOARD DESIGN

Independent

- Designed a Macropad layout using Kicad and EasyEDA
- Key features include USB C PD, rotary encoder, MX-compatible switches
- Worked with QMK firmware and flash via ISP
- Communicated with vendors to get cost effective BOM (PCBWay)

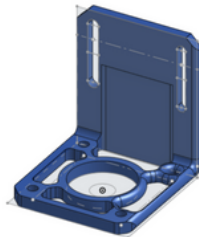


July 2024 - August 2024

Stepper Motor MicroStepping

Independent

- Designed motor mount in Onshape for Stepper motor control
- Used TB6600 Stepper Motor driver for microstepping and AS5600 for angle detection
- Wrote Arduino code for sensing
- Optimized shape of parts via Pocketing code

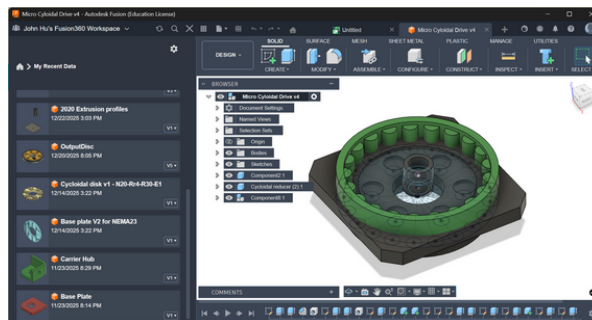


December 2026 - Present

Micro Cycloidal Drive

Independent

- Worked with Fusion360 to design eccentric shaft, cycloidal gears, and housings
- 3D printed basic parts
- Ongoing WIP



Tools

- Fusion 360
- Ansys Fluent
- Onshape
- KiCAD
- C++
- MATLAB/Simulink
- Arduino
- Cura/Prusa Slicing

References

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